Roni Saguey

AI Systems Engineer — agent infrastructure, LLM tooling, applied computer vision

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SUMMARY

Self-taught engineer building production agent infrastructure: LLM API routers, context-compression systems, agent-memory engines, and computer-vision pipelines. Ship end-to-end — architecture, implementation, tests, and deployment — across Python, JavaScript and shell. Multiple competition entries in applied AI, including national hackathons in computer vision and agricultural ML.

SELECTED PROJECTS

Webchat-to-API Harness ★ 9

Open source

`JavaScript · Node · Puppeteer · OpenAI + Anthropic API schemas`

- OpenAI- *and* Anthropic-compatible HTTP API backed by a real webchat tab, so any agentic client (Aider, LiteLLM, Claude Code via `ANTHROPIC_BASE_URL`) can drive a session you own.
- Full tool-call support — file read/write and bash execution surfaced through the chat UI.

free-ai — Multi-Provider LLM Router ★ 2

Open source

`JavaScript · zero runtime dependencies`

- One local endpoint in front of every free LLM tier: ranked model list, automatic fallback, and a linear per-model cooldown. Serves 50+ models across 10 providers.
- Zero dependencies — deliberately dependency-free so it runs anywhere with a Node binary.

AxiomMem — Agent Memory Engine HACKATHON ENTRY

Agent Memory Challenge 2026

`Python · FastAPI · SQLite (WAL) · embeddings`

- High-performance memory system engineered for execution continuity, temporal resolution and zero-loss retrieval across coding and textual-memory tracks.
- Built and verified end-to-end against the live competition API: contract compliance, tenant isolation, idempotency, and large-scale ingestion.

Agentic Visual Inspector COMPETITION ENTRY

OpenCV AI Competition 2026

`Python · OpenCV 5 · YOLOv10 ONNX · LLM reasoning loop`

- Vision system that does not just detect a defect — it investigates it against the line's recent history and recommends a supervisor-readable action.
- Runs real-time detection on CPU (YOLOv10n via `cv2.dnn`) with a reasoning layer that weighs an isolated blemish against a repeated fault trend.

commit-condom — Push Gate for Coding Agents ★ 1

Open source

`JavaScript · git hooks · proxy architecture`

- Policy gate that inspects every commit for size, file count and secret patterns *before* it reaches a remote; the agent holds a placeholder credential and the proxy holds the real one.
- Found and fixed a silent failure where every default secret pattern was skipped — a gate that inspected nothing and passed everything.

DriftClean — LLM Session Sanitizer ★ 2

Open source

`Python · streaming interception · pluggable judge models`

- Intercepts a model's reasoning stream, scores it for refusal / fake compliance / off-task drift, and asks a second judge model for a verdict before emitting a drift event.
- Works with any provider that streams thinking tokens; zero dependencies.

LabLens — Bloodwork Interpreter

UnivaBio · DSH Hacks

`Python · OCR · reference-range grounding`

- Photograph a lab report; it reads the values, flags out-of-range results, and explains each in plain English — grounded strictly in reference ranges, never a diagnosis.

CornShield AI — Crop Disease Triage

4-H AI in Agriculture 2026

`Python · Flask · image classification · 21 passing tests`

- Foliar disease early-detection and severity triage for Midwestern corn growers, with extension-sourced treatment guidance. 21/21 tests passing; zero secrets committed.

Tool-Call Compactor — Context Compression

Open source

`JavaScript · MCP · token accounting`

- Compacts tool-definition payloads in a system prompt to cut token cost, exposing 1,100+ tools through a searchable index without loading their schemas.

TECHNICAL SKILLS

Languages	Python, JavaScript / Node.js, Bash, SQL, HTML/CSS
AI / ML	LLM routing & fallback, agent memory systems, prompt/context compression, computer vision (OpenCV, YOLO ONNX), OCR pipelines, embeddings
Infrastructure	Linux, systemd, Docker, Cloudflare tunnels, SQLite (WAL), REST APIs, git
Integrations	OpenAI & Anthropic API schemas, MCP servers, Puppeteer/CDP browser automation, FastAPI, Flask

COMPETITIONS & RECOGNITION

- **OpenCV AI Competition 2026** — submitted entry (Agentic Visual Inspector).
- **Agent Memory Challenge 2026** — built and live-verified a competition-grade memory engine (AxiomMem).
- **4-H AI in Agriculture 2026** — division entry, CornShield AI.
- **ForgeHacks Online 2026** — submitted entry (ClarityCheck).

All projects above are public at github.com/ronisaguey-ux. Self-taught; everything listed was designed, implemented and tested by hand.